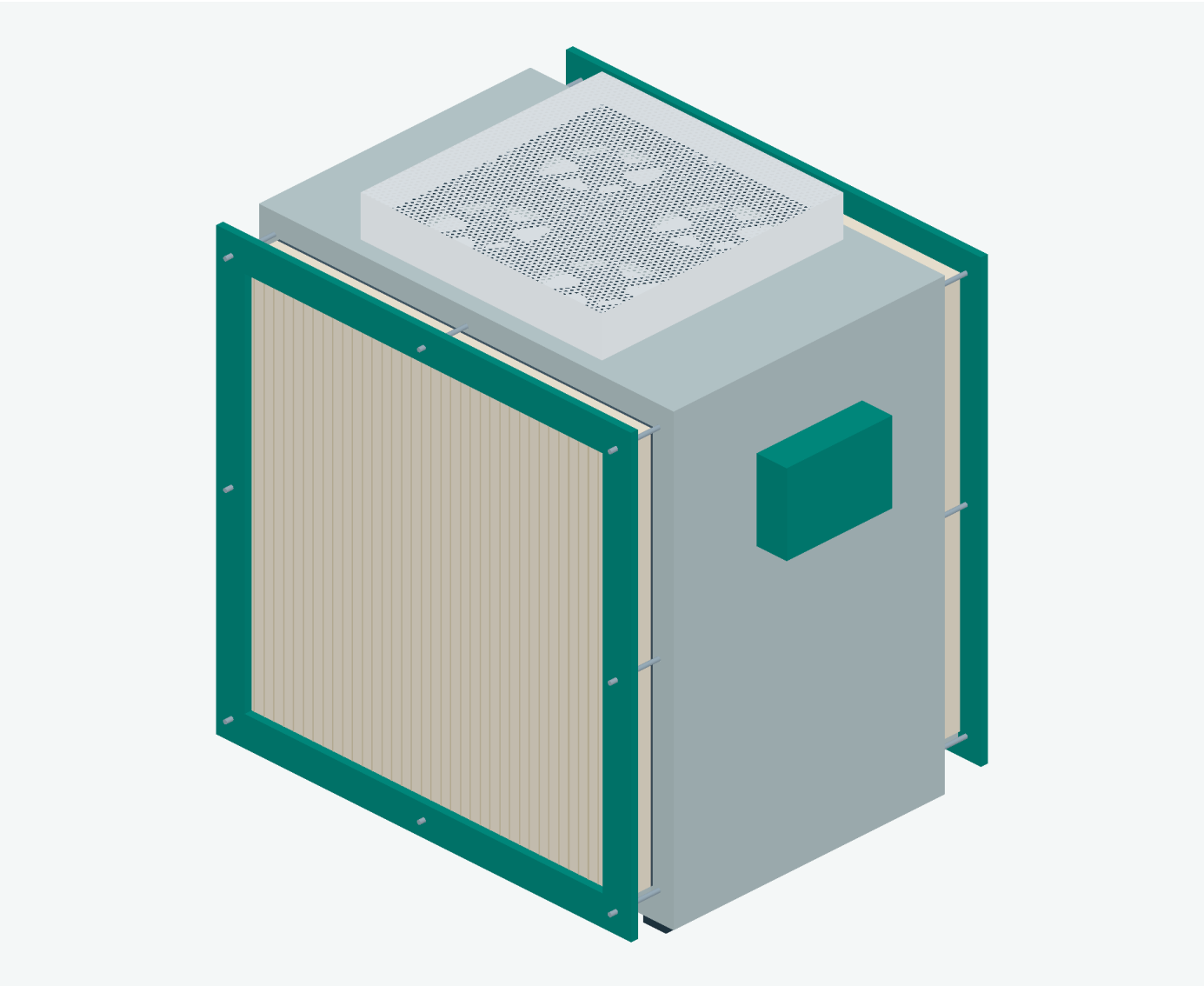


A prototype we can manufacture

An indoor particle-filtering demonstrator, built around our own housing, seals, controls and test method.



Two large filter panels pull particles out of incoming air. Four small fans move that air upward. A sealed, serviceable box lets us measure whether the assembly actually works.

Proposed build	What exists now
550 mm wide cabinet; 631 mm overall height; about 473 mm filter-to-filter depth. External controls add 40 mm to width.	Native CAD, exported STEP, dimensioned panel drawings, pressure calculations and a connector-level electrical plan.
First goal: a measured indoor result, not a large outdoor claim.	Physical prototypes built: 0. No parts ordered. No physical results.

RECOMMENDATION: fund one small demonstrator before the full cylindrical tower. The final outdoor goal remains; this is not a replacement claim or a purchased purifier presented as our invention.

What changed and why

The immediate work is one assembly, not another cosmetic tower branch.

Usable previous work	Reason it is not a build release
Airflow / filter studies, FreeCAD scripts and pressure-budget logic.	Historical 1,332 m3/h CFD was partial/nonconverged; it does not validate this new assembly.
Measurement recorder and indoor decay-analysis scripts.	Synthetic software tests only. No instruments or measured air-cleaning result.
Cylindrical packaging and structural studies.	Heavy casing, unresolved fan duty, real filter fit and closure details stalled integration.

The faster route. Flat cut panels replace curved fabrication for this demonstrator. An external OEM power supply avoids designing an exposed mains power cabinet. The two filters share airflow in parallel, reducing the resistance each fan must overcome.

What D01 can validate. Assembly method, replaceable seals, delivered airflow, electrical consumption, noise and repeatable indoor particle-decay measurements. MERV13 filters are NOT HEPA; D01 does not prove the old H13 stage, gas removal, water stage, outdoor bubble size or tower spacing.

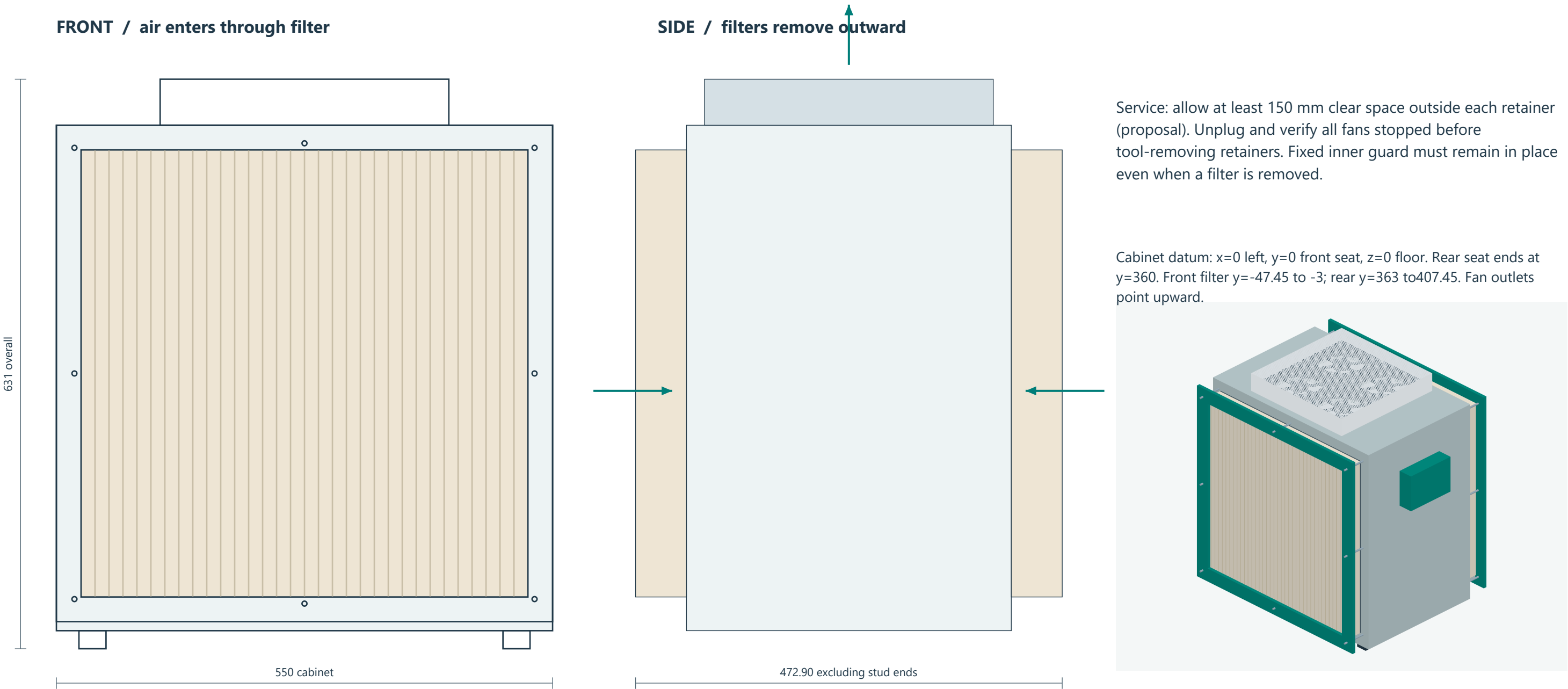
Funding request. Obtain quotations for the BOM, panel cutting, guard fabrication, assembly review, rental or paid airflow/pressure/power/noise measurement, and particle monitors. Total cost, freight, taxes and lead times are UNKNOWN. No guessed purchase total is presented.

Critical path	Owner / dependency
1. Resolve exact fan revision and obtainable filter; confirm sizes and filter curve.	Digital selection update + vendor evidence. No contact made.
2. Complete guard attachments, joints, seal and control safety review; issue R01 release.	Our drawings + paid mechanical/electrical review. No college lab assumed.
3. Purchase, cut panels, assemble and inspect.	User authorization; supplier delivery and fabricator schedule unknown.
4. Commission and run repeatable indoor trials.	Qualified physical checks + rented/bought instruments.
Next measurable milestone	One guarded, leak-checked operating assembly with measured total flow, watts and recorded test conditions.

Digital engineering can proceed now. Delivery, fabrication and testing cannot be assigned honest completion dates until suppliers and services confirm them. No outdoor scale-up is on this release path.

M00 General arrangement

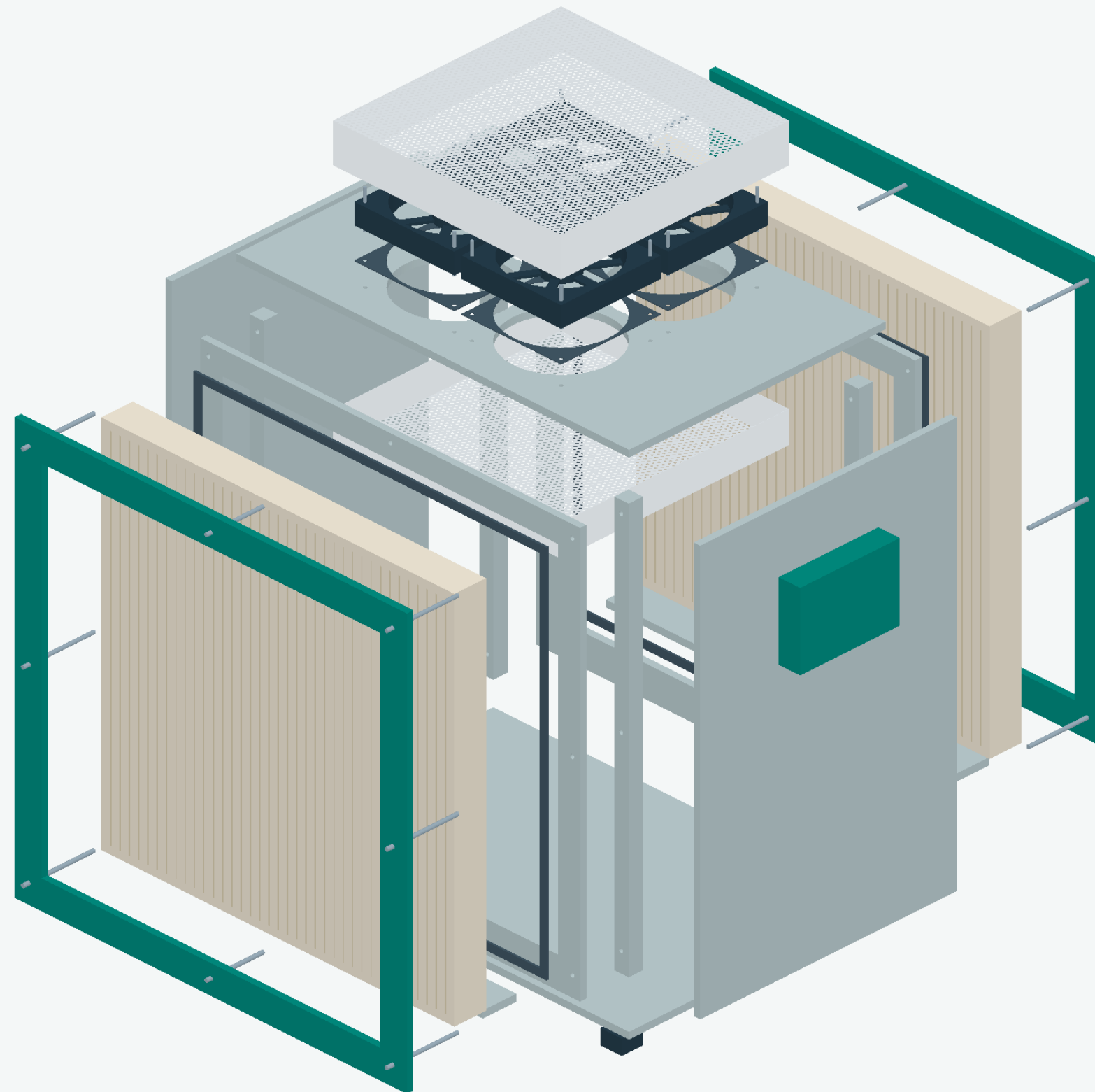
Dimensions in mm. Proposed geometry; views not to scale. Guards are sheet envelopes, not certified perforation geometry.



Interface	Nominal geometry / evidence
Filter support and seal	495.3 square x44.45 deep OEM envelope; 470 square intake; 12.65 mm support overlap/edge. 10 mm seal width needs actual frame landing check.
Fan plate	Four140 square fans; centres(200,105),(350,105),(200,255),(350,255); 125 square mounting pitch from OEM.
Service clamping	550 square retainer,475 opening,8 M5 studs per face; shelf supports filter weight. Foam4mm free to3mm installed is an assumption.
Structure	9mm plywood proposal with20x20 internal corner cleats and non-slip feet. Not a certified load or stability design.

M00 Exploded assembly

Same CAD parts translated for assembly explanation. The exploded view is not another design option.



1 Fan cassette

Four P14 Max envelopes, gaskets and through-bolts sit on a drilled top plate. Outer guard encloses rotating parts.

2 Cabinet

Base, two end panels and two aperture seats join against internal cleats. Seal every fixed seam; gasket the removable top.

3 Filter service

Each filter sits on its shelf, against a continuous gasket. A removable retainer spreads clamping around the paper frame.

4 Protection

A tool-fixed inner guard stops access to fan blades through a removed filter. Guard perforations and attachment strength still require detailed review.

5 External controls

Control enclosure is a space reservation only. Power supply stays external; cable route, gland and actual mounting need completion.

CAD simplifications: fan rotor is illustrative; filters are envelopes; guard sheets do not contain their perforation holes. Nuts, washers, cleat screws and guard flanges are not fully modeled. These exclusions prevent an honest fabrication release today.

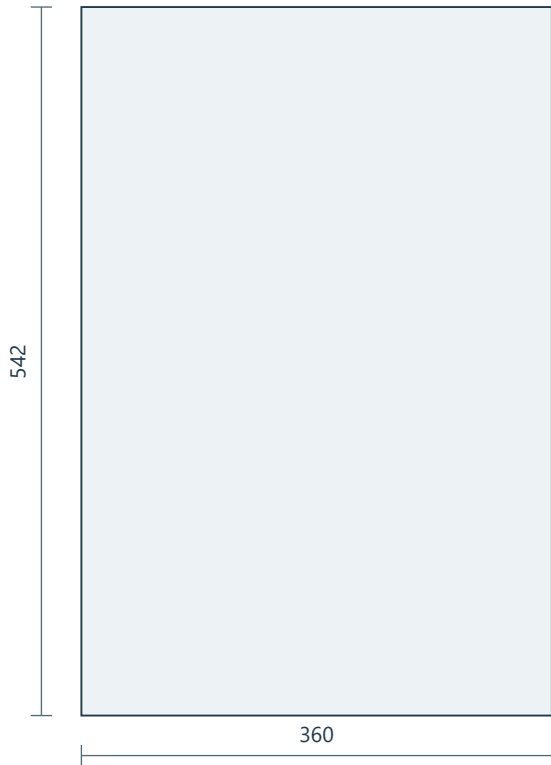
M01 to M04 Panel drawings

All dimensions mm. DXFs are 1:1 review geometry, without kerf compensation. Do not scale the PDF to cut parts.

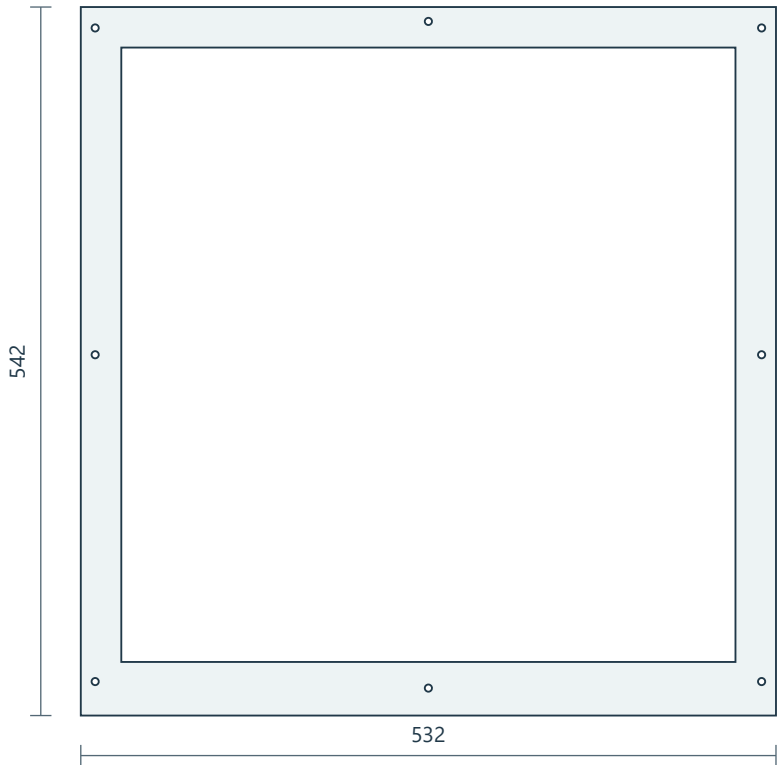
M01 BASE / qty 1 / t=9



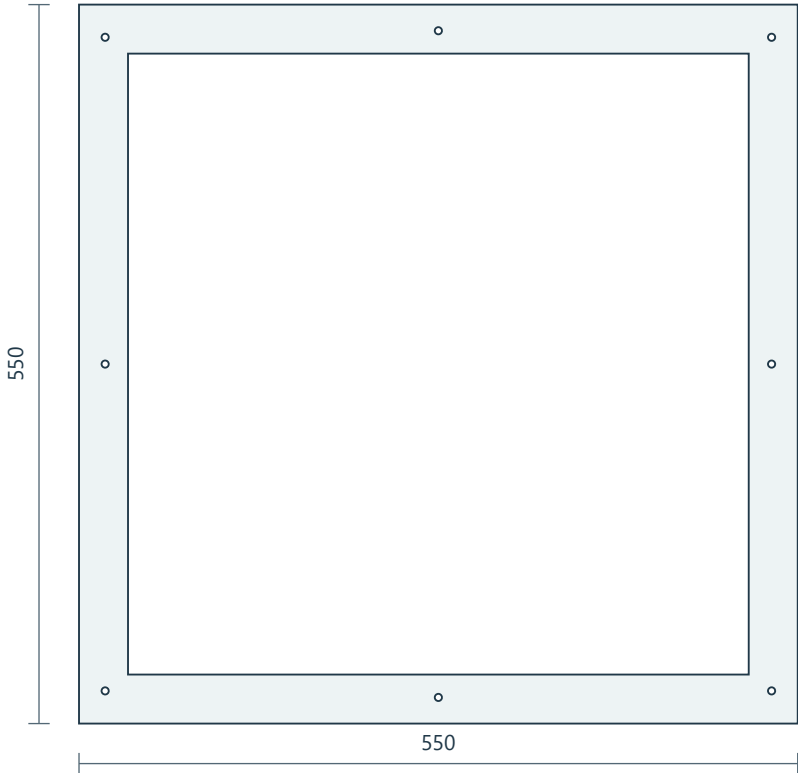
M02 END / qty 2 / t=9



M03 SEAT / qty 2 / t=9



M04 RETAINER / qty 2 / t=9



Hole coordinates for M03 seat
Datum lower left of 532 x 542 panel. Aperture lower left (31, 41), 470 x 470.
8 holes dia 5.5: x=11 or 521 at z=26, 276, 526;
plus (266, 21), (266, 531).

Hole coordinates for M04 retainer
Datum lower left of 550 x 550 panel. Aperture lower left (37.5, 37.5), 475 x 475.
8 holes dia 5.5: x=20 or 530 at z=25, 275, 525;
plus (275, 20), (275, 530). Minimum centre-to-edge distance 20mm on retainer. Use flat washers; verify bending and plywood bearing.

M05 SHELF / qty 4 / t=9



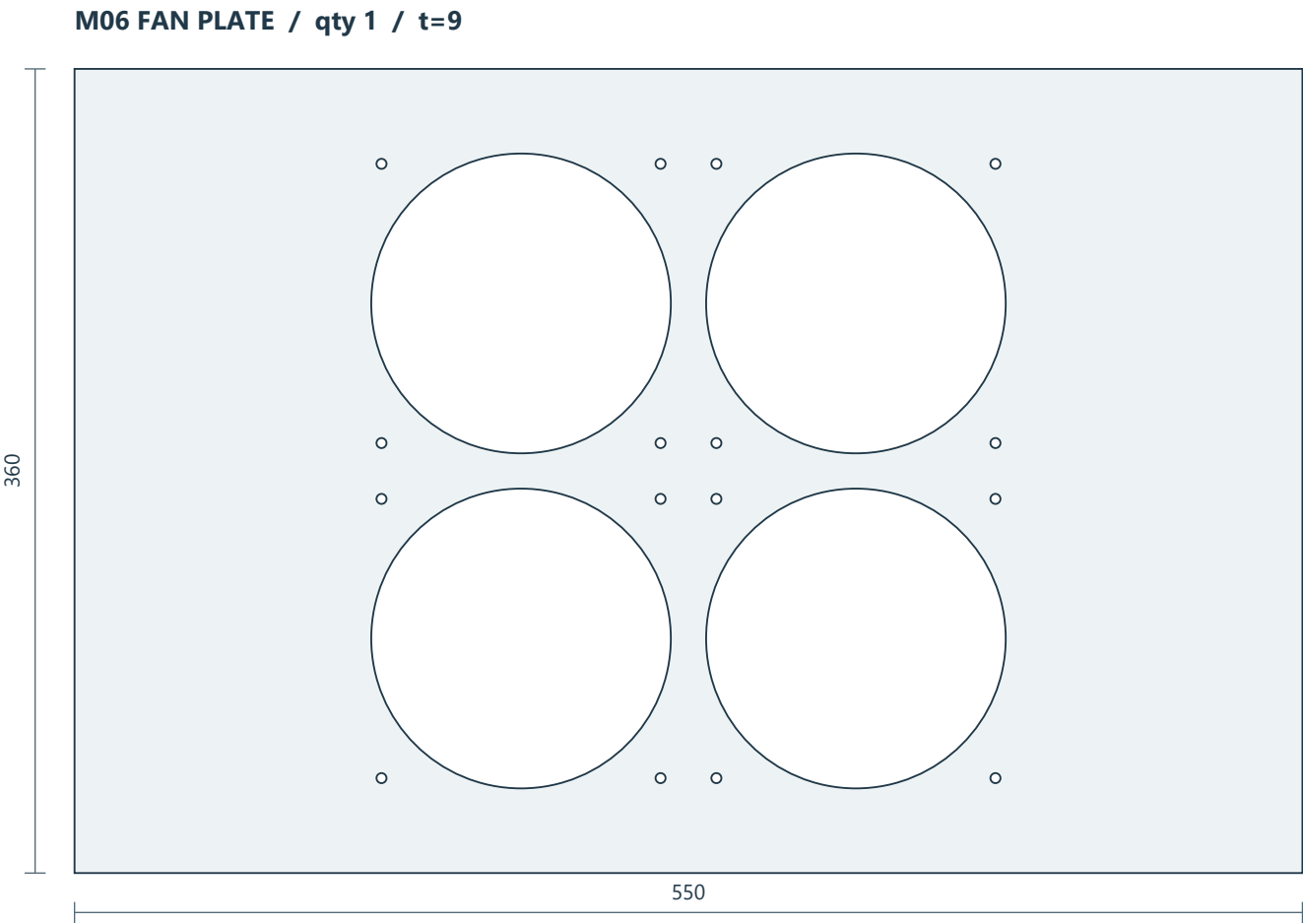
Two ledges per filter: x30 to 265 and 285 to 520. A 20mm centre gap clears the lower clamp stud. Top surface at z57.35. Match-drill corner-cleat stud holes from the seat; short centre studs do not pass through cleats.

Proposed material and joints
9mm sealed plywood, 20x20 timber cleats. Pilot-drill; screw into cleats rather than panel edges. Adhesive/sealant must be compatible, fully cured and suitable for indoor exposure. Target panel dimensions +/- 0.5mm and squareness within 1mm are proposed, not achieved tolerances.

The retainer was widened to 550mm to fix weak edge ligaments found during review. M5x100 studs now reach through the corner cleats. Cleat screw pitch, pull-out capacity, guard brackets and retainer stiffness still require completion.

M06 Fan plate and S01 filter seal

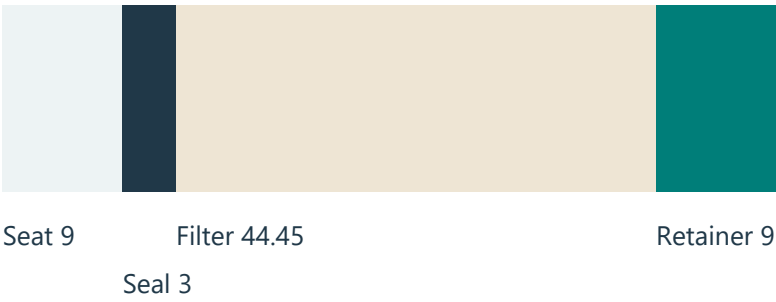
OEM fan dimensions establish the pattern. Filter sealing and clamping require physical fit confirmation.



Fan centres from lower-left datum:
(200,105), (350,105), (200,255), (350,255).
Each fan: aperture dia134.2; four dia4.5 holes at centre +/-62.5 in x/y. Fan body140 x140 x27.
Verify the delivered revision with the OEM1:1 template.

Proposed mounting: M4x50 through-bolts, flat washers and locknuts,16 sets. Do not over-compress the fan frame. A1mm installed fan gasket is assumed. Seal around the cable entry; do not trap cables in the rotor or plate joint.

Filter edge section / schematic, not to scale



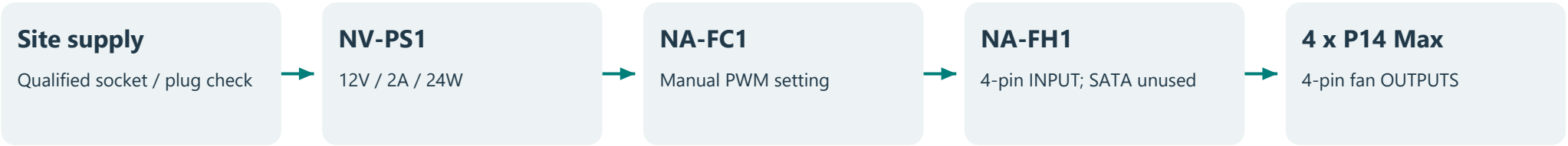
Clamp force comes from retained studs/fasteners. Do not use fan suction as the seal.
Shelf takes gravity load. No adhesive on replaceable media.

Seal landing: proposed10mm wide closed-cell foam around the495.3mm frame; butt joint sealed or use a continuous ring. Free4mm / installed3mm is only a starting compression assumption. Verify manufacturer force data, flatness, frame width and bypass before specifying torque or stops.

Guard proposal: two320 x320 perforated faces,4mm holes on6mm triangular pitch (~40% open), with solid skirts. CAD shows sheet envelopes. Inner face z530, outer face z630. Final flanges, fixing holes, edge safety, blade clearance and finger-probe assessment must be detailed before release.

E01 Power and control

Connector-level preliminary schematic. OEM preterminated cables only; no custom mains wiring.



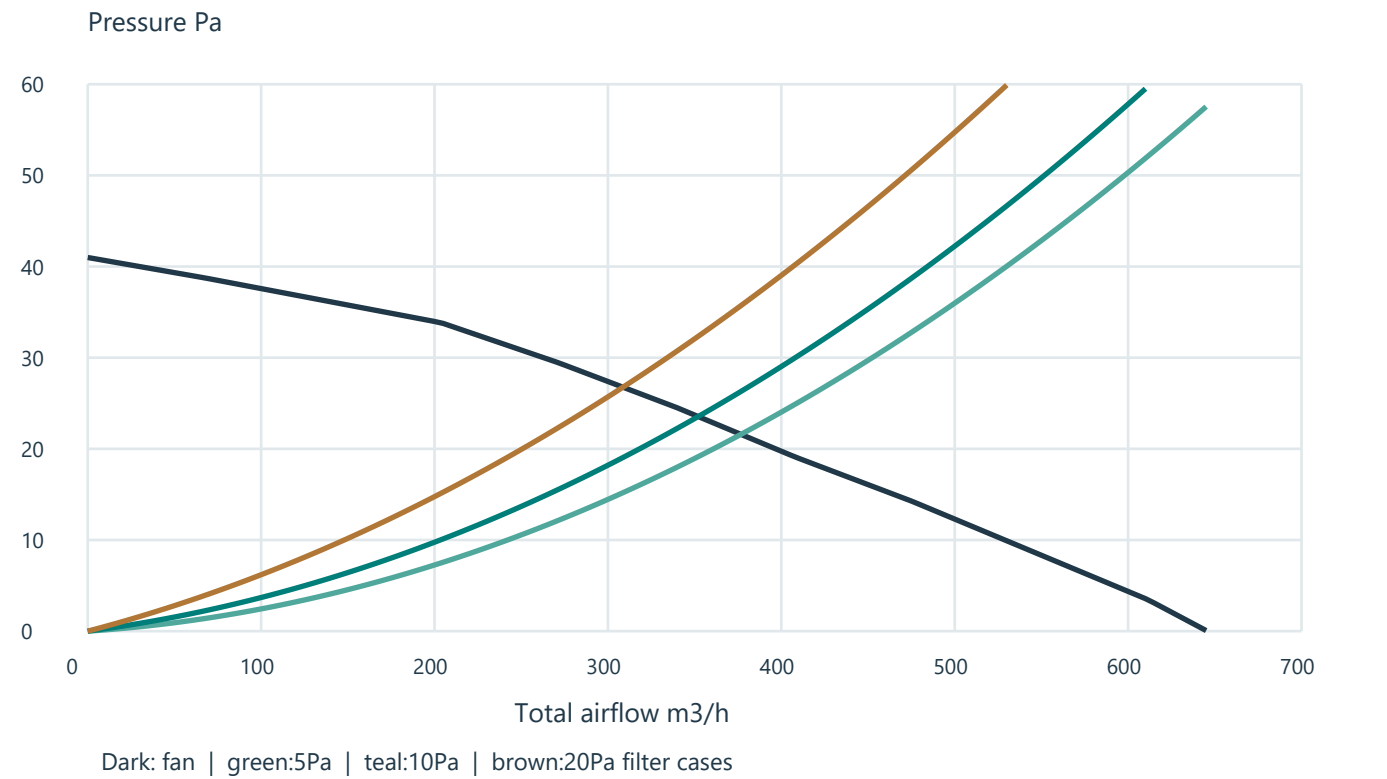
NV-PS1 -> included NA-AC10 -> NA-FC1 input -> NA-FH1 PWM input -> ports1-4

Verified OEM basis	Design implication / unresolved check
NV-PS1:12V,2A,24W; external ClassII adapter. OEM supports FC1/FH1 combination.	Keep supply closed and outside cabinet. Supplied EU/US/UK plugs do not establish compatibility with the actual Indian socket.
NA-FH1:24W via4-pin input; eight outputs. NA-FC1 total current limit3A.	Use only4 fans. No SATA feed, no motherboard, no double powering. Route cables outside blade paths with strain relief.
Fan nominal current4 x0.35A=1.4A; nominal fan load16.8W.	7.2W margin before controller/hub losses and startup. Not a measured maximum or proven start-current budget.
Keyed OEM4-pin connectors; fan drawing identifies ground/power/tach/PWM.	No bare-terminal assignment or connector orientation invented. Confirm delivered manuals and third-party PWM compatibility before energizing.

RELEASE GAP: This chain may restart when power returns. It DOES NOT implement the existing tower manual-restart/interlock requirement. Do not silently waive that requirement. Before supervised testing, a qualified reviewer must either implement a rated restart-inhibit arrangement or explicitly approve a guarded bench-only exception with unplug-before-service controls. No unattended use. Speed=0 is not isolation.

The pressure budget

Screening scenarios at maximum fan speed. These are calculations, not a measured airflow rating.



At each calculated operating point	Low clean	Middle clean	Loaded case
Flow m3/h	376.7	352.1	308.7
Fan pressure Pa	21.6	23.5	26.8
Filter path Pa	4.71	8.80	15.43
Intake/plenum Pa	3.08	2.69	2.07
Inner guard Pa	5.87	5.13	3.94
Outer guard Pa	5.87	5.13	3.94
Installation reserve Pa	2.05	1.79	1.38

Four fans share the flow; their pressures do not add. Two filters each see half the flow; their pressure drops do not add. Assumed filter loss is 5/10/20Pa at total 400m³/h, scaling linearly. Other losses use $K \times \rho \times v^2 / 2$. Exact filter curve and installed guard losses remain UNKNOWN.

The fan curve is visually digitized from the OEM chart. The +/-2Pa sensitivity gives about +/-11m³/h, but this is NOT a total uncertainty bound. The installation reserve is an allowance, not a verified static/total-pressure conversion. No CADR or cleaning percentage follows from these numbers.

Parts and funding schedule

Exact OEM references are separated from proposed fabrication items. All prices and delivery dates remain unquoted.

ID	Qty	Product / fabrication item	Technical basis	Status before purchase
F01	2	S&P; MERV13 990303 reference	495.3x495.3x44.45mm OEM	Dimensions verified; India availability and resistance curve UNKNOWN; not HEPA
F02	4	ARCTIC P14 Max	140x140x27;125mm pitch;12V0.35A	OEM verified; SKU/color revision conflict. Checked India listings out of stock
E01	1	Noctua NV-PS1 + included NA-AC10	12V2A24W external supply	OEM verified; India stock, plug compatibility and price UNKNOWN
E02	1	Noctua NA-FC1	Manual PWM,3A maximum total	OEM verified; third-party fan behavior / current / availability to confirm
E03	1	Noctua NA-FH1	24W via4-pin input	OEM verified; use4 outputs only; included input cable
M01-06	12	Cut9mm plywood panels	Base1,end2,seat2,retainer2,shelf4,top1	Proposed; drawings/DXF supplied; material, joints and clamp strength unreleased
M07	4	20x20 cleats,542mm long	Corner joinery support	Proposed; screw schedule and load review outstanding
S01	~4m	10mm wide closed-cell foam	4mm free /3mm installed proposal	Exact product, force-compression, frame landing and leakage unknown
S02	4	140mm fan gaskets	1mm installed proposal	Exact material / apertures / compression unknown
H01	16 sets	12 M5x100 +4 M5x80 studs/nuts/washers	6 long side /2 short centre studs per face	Proposed; short centre studs avoid inner guard. Thread engagement / retention review needed
H02	16 sets	M4x50 bolts/washers/locknuts	Fan mounting	Proposed; actual stack and rotor clearance to verify
G01-02	2 cages	Perforated guards + solid skirts	320x320 faces; outer50 / inner41 high	4mm holes6mm triangular pitch proposal; attachment and reach review open
M08	4	Non-slip feet	30x30x20 envelopes	Exact product / attachment / tip and slide checks unknown
E04	1 set	Control enclosure/cable fittings	Reserved140x100x40 external space	Unselected; actual components and wiring layout not detailed
H03	1 lot	Cleat screws, guard brackets, seam sealant, top gasket	Shop detail pending	NOT omitted from cost; quantities and products unresolved
T01	1 service	Airflow / pressure / watt / sound measurement	Rent or hire equipped operator	No college lab assumed; quote and instrument suitability needed
T02	2	Logging particle monitors + temp/RH	Co-locate before trials	Exact kit / calibration or comparison plan / quote pending
R01	1 service	Mechanical + electrical review / commissioning	Defined in final page	Scope-specific paid service; no named engineer assumed

Assembly and physical trial

Work to the released revision only. This R00 supports design review and quotation, not immediate cutting.

1 Receive and measure

Check actual fan identity/pitch, filter size and edge frame, power accessories and manuals. Measure with calipers; record photos and serials. If they differ, update the CAD rather than force fitting.

2 Make the cabinet

After joint/guard review: cut, deburr and seal panel edges. Square base/end/seat panels on cleats. Pilot-drill and attach the supports. Keep adhesive away from filter media; allow full specified cure.

3 Fit the service parts

Install filter shelves and continuous gaskets. Seat filters with arrows inward. Fit retainers and retained fasteners, tighten evenly without crushing frames. Fit the gasketed removable fan plate and four outward-blowing fans.

4 Fit and check guards and controls

Secure inner and outer guards with tool-required fasteners; verify no reachable blades, loose edges or blade contact. Mount controls and secure cables. Qualified person resolves restart behavior, socket/adaptor, protection and startup current before energizing.

5 Commission without pollutant generation

In a clean supervised room, measure cold start/current, all fan rotation, low/high settings, power-loss response and at least 30 min high-speed temperature/noise behavior. Stop on rubbing, smell, instability, loose guards or abnormal heating. This trial is not a thermal safety certification.

6 Measure airflow and seals

Use a calibrated low-pressure instrument (roughly 0-100 Pa) and a competent low-loss flow hood/traverse method. A restrictive hood changes these low-pressure fans: document its correction/backpressure. Check all seams and compare with temporarily taped perimeter seams. A significant flow/particle change requires sealing repair; no visual seal claim alone.

7 Run matched indoor trials

Record room volume, doors/windows, background conditions, monitor locations, fan setting, watts and filter pressure. Co-locate two monitors first. Run at least 3 matched fan-off/fan-on decay pairs with comparable starting conditions, no occupants/pollutant sources and documented ventilation. Do not generate smoke, burn incense or handle test dust casually. If ambient particles are too low, book a controlled test service.

8 Report what actually happened

Preserve raw files; use existing recorder/decay analysis with uncertainty and natural-decay comparison. Report measured flow, power, noise, repeatability and particle response. A low-cost PM sensor cannot certify MERV/HEPA efficiency. Negative or inconclusive results are valid results.

Evidence and the release boundary

What was checked, what is still missing, and what the next revision must close.

Evidence class	Current result
DETECTED digital checks	73 valid CAD objects; 435 enclosure/filter/support pairs and 1,280 fastener-to-part pairs have no positive-volume clashes. Nine arithmetic tests pass. Not strength, leakage or safety certification.
DETECTED and corrected	Retainer widened to 550 mm; minimum bolt edge distance 20 mm. Side studs reach cleats; shorter centre studs clear guard. Shelf split clears bottom stud. Guard flanges/joints and cleat fastening remain incomplete.
VERIFIED OEM facts	Fan envelope/pitch, filter reference dimensions and power/control product ratings have source links below. No exact procurement set is released.
ASSUMPTIONS	Panel/cleat material, seals/compression, guard loss, filter pressure scenarios and flow sharing. Hardware masses and structural capacity incomplete.
UNKNOWN physical results	Airflow, sealing, efficiency, CADR, noise, startup current, stability, surface temperatures, outdoor benefit. Physical prototypes: 0.

Qualified review has a specific job: check panel/cleat joints, retainer bending and edge distances, shelf loads, actual mass/centre of gravity, tip/slide resistance, guard reach/deflection and cable restraint. Electrical review checks the OEM chain, actual plug/socket, startup budget, automatic-restart gap and safe isolation. Commissioning verifies these on the built unit. This is not a request to redesign everything from scratch.

Public build precedent: Tempest Brisa provides DXF/SVG construction files under CC0 1.0. One DXF and its licence are archived in data/prototype_d01/sources. The D01 enclosure is newly generated, not a rebranded Brisa or Dyson model. No upstream CADR result is inherited and no patentability claim is made.

Sources checked 03 October 2026

[ARCTIC drawing and mounting template](#)

[ARCTIC specification and pressure chart](#)

[S&P MERV13 dimensional submittal](#)

[Noctua power chain](#)

[Noctua hub ratings](#)

[Noctua controller current limit](#)

[Brisa source and CC0 licence](#)

[India listing checked: out of stock](#)

Clickable source titles. OEM files are reference material, not open-source manufacturing licences. Availability can change; no supplier has confirmed stock, suitability, a quote or a date to this project.